Stat Analysis & Algebraic Reasoning — Unit 1: Asking Questions with Data: Study Design

Practice Test — Study Copy— Answer Key

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. **100 points:** Part A 20, Part B 20, Part C 40, Part D 20.

Part A — Vocabulary — 10 questions, 2 points each

Write the letter of the matching description next to each term.

1.	**D**	Population	A. A number that describes a sample.
2.	**H**	Sample	B. The researcher assigns people to groups and gives them a treatment.
3.	**G**	Parameter	C. A variable whose values are labels, like a color or a team name.
4.	**A**	Statistic	D. The whole group we want to learn about.
5.	**C**	Categorical variable	E. Everyone has an equal chance of being picked.
6.	**I**	Quantitative variable	F. The researcher watches or asks, without assigning anyone to a group.
7.	**E**	Simple random sample	G. A number that describes a whole population.
8.	**J**	Convenience sample	H. The part of the population we actually collect data from.
9.	**F**	Observational study	I. A variable whose values are measured amounts, like minutes or inches.
10.	**B**	Experiment	J. Choosing whoever is easiest to reach.

Part B — Multiple Choice — 5 questions, 4 points each

11. Which of these variables is **quantitative**?

    (A) Eye color — Letter: C
    (B) Favorite sport
    (C) Height in inches ← **correct**
    (D) Home ZIP code

12. Maplewood High School has 900 students. A survey of 60 of them finds that 30% walk to school. The 30% is a

    (A) statistic, because it describes the sample. ← **correct** — Letter: A
    (B) parameter, because it describes the sample.
    (C) statistic, because it describes all 900 students.
    (D) parameter, because it describes all 900 students.

13. A librarian takes an alphabetical list of card holders, picks a random start, and then chooses **every 10th name**. This is a

    (A) stratified sample. — Letter: D
    (B) cluster sample.
    (C) convenience sample.
    (D) systematic sample. ← **correct**

14. To learn how often town residents exercise, a researcher surveys people **inside a gym**. The results will most likely

(A) be about right, because the people were chosen fairly.
(B) overestimate how often residents exercise. ← **correct**
(C) underestimate how often residents exercise.
(D) be right if the researcher surveys more people at the gym.

Letter: B

15. Researchers use a coin flip to put 80 volunteers into two groups. One group sleeps 8 hours and the other sleeps 5 hours before a memory test. This study is

(A) an experiment, because the researchers assigned the groups. ← **correct**
(B) an experiment, because it measures memory.
(C) an observational study, because the volunteers were watched.
(D) an observational study, because there are two groups.

Letter: A

Part C — Short Computations — 4 questions, 10 points each

16. **Silver Lake Recreation Center** has 900 members. The director asks 60 members chosen at random, and 18 of them use the pool every week.

(a) What percent of the sample uses the pool every week? **30%**

$$\frac{18}{60} = 0.30 = 30\%$$

(b) About how many of all 900 members use the pool every week? **270**

$$0.30 \times 900 = 270 \text{ members}$$

17. **Granite Falls Middle School** has 600 students and wants a **stratified** sample of 60 — the same fraction from each grade.

(a) What fraction of the school is being sampled? **1/10**

$$\frac{60}{600} = \frac{1}{10}$$

(b) Complete the table.

Grade	6	7	8	Total
Students in grade	240	200	160	600
Number sampled	24	20	16	60

18. **Brookside Apartments** has 500 households on a numbered list. The manager wants a **systematic** sample of 50.

(a) Find the interval k. **10**

$$k = 500 \div 50 = 10$$

(b) The random start is household 3. List the first four households chosen.

3, 13, 23, 33

19. In an **experiment**, 40 students are randomly assigned to use a flashcard app and 40 to study the usual way. Afterwards, 30 of the app group and 20 of the usual group passed a quiz.

(a) Find the percent who passed in each group. 75% 50%

App: $\dfrac{30}{40} = 0.75 = 75\%$

Usual: $\dfrac{20}{40} = 0.50 = 50\%$

(b) How many percentage points apart are the two groups? 25 points

Part D — Critical Thinking — 1 question, 20 points

20. A random sample of Maplewood students found that students who **eat breakfast** have higher grades than students who skip it. Nobody was assigned — students chose for themselves. The principal says, *"Eating breakfast causes higher grades."*

Is the principal's conclusion justified? Explain your reasoning, and name one **other variable** that could explain the result.

No. Nobody was assigned — students chose — so this is an observational study. It shows an association between breakfast and grades, not causation. Another variable: family routine. Students with more support at home are more likely to eat breakfast *and* to keep up with homework, and that support could raise grades by itself. Only an experiment that randomly assigns breakfast could show it causes higher grades.